

AYUSH KUMAR SINGH

Software Engineer | Backend & Full-Stack | MERN | REST APIs | Real-Time Systems

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EDUCATION

TECHNO MAIN SALT LAKE

Salt Lake, Kolkata

B. Tech in Computer Science and Business Systems

2022 – 2026

Senior Secondary Education (ISC)

2021–2022

Secondary Education (CBSE)

2019–2020

TECHNICAL SKILLS

- Languages: C++, Python
- Web Development: HTML, CSS, JavaScript, React.js, Node.js, Express.js
- Databases: MySQL, MongoDB
- Tools & Technologies: Git, GitHub, Postman, REST APIs, n8n AI Tools
- Core Concepts: Data Structures & Algorithms, OOP, Computer Networks

PROJECTS

Just In Time – Resilient Proxy-Based Testing Platform | 2025 | [GitHub Repository](#)

Tech Stack: Node.js, Express.js, React, n8n, GenAI (Gemini)

- Designed and implemented an **intelligent proxy layer** to ensure zero-downtime by automatically rerouting traffic to fallback services during backend failures.
- Built a **self-healing reliability mechanism** using an AI watchdog that monitors error rates and triggers a kill switch to stabilize the system.
- Developed a **logic-aware testing workflow** where business requirements written in plain English are validated against **backend services and code** using AI-driven automation.

Collabsuite – Real-Time Collaboration Platform (In Progress) | [GitHub Repository](#)

Tech Stack: MERN, Socket.io, WebRTC

- Designing and implementing a real-time collaboration platform with live document editing, whiteboard, chat, and task management.
- Built core real-time communication using **Socket.io** and **WebRTC** signaling for multi-user collaboration.
- Focused on minimizing latency and synchronization issues in multi-user real-time interactions

Quick.ai – AI SaaS Platform | 2025 | [Live Demo](#) | [GitHub Repository](#)

Tech Stack: MERN, OpenAI API, Cloudinary

- Developed a **full-stack AI SaaS platform** enabling content generation, image creation, and resume analysis using large language models.
- Architected **modular backend services** with Cloudinary-based image processing to ensure performance and scalability.
- Implemented authentication and subscription management for a **live SaaS platform supporting 200+ users**, along with a responsive UI to ensure secure and scalable user experience.

INTERNSHIP

Web Development Internship (Remote) | 2025 | [Link](#)

- Built responsive React applications and integrated REST APIs.
- Used **Git/GitHub** for version control and performed **debugging and code optimization**.

ACHIEVEMENTS

Smart India Hackathon 2024 — Selected in first round | [Certificate](#)

Calcutta Hacks 2025 — First Runner-Up | [Certificate](#)